Looking at real wood anatomy

A supplementary guide for Lesson 2. Open the linked resources and examine the cells and growth-ring features visible in transverse sections. You do not need previous experience identifying wood species.

Compare two kinds of wood

Open the Forestry and Forest Products Research Institute (FFPRI) micrograph page. It shows electron micrographs of Japanese cedar, a softwood, and mizunara oak, a hardwood. The image contributor is credited on the page. The page text is in Japanese.

1. In cedar, observe the arrangement and apparent sizes of the cells.

2. In oak, look for the distribution of comparatively large pores.

3. Before comparing images, check the magnification, plane of section and specimen information.

Compare additional species

Satoshi Nagai's wood electron micrograph resource (Hyogo Prefectural Technology Center, PDF, Japanese) presents images and anatomical descriptions for several species, including oak, chestnut, beech and linden. Select examples of ring-porous and diffuse-porous wood and compare the sizes and distribution of vessels. The PDF is large and may take time to load.

The FFPRI research databases page also links to databases containing Japanese wood anatomy images. Availability depends on the external service. Both pages are in Japanese, but the photographs can be examined alongside this guide.

For specimen-specific optical micrographs, try Picea glehnii, Quercus crispula, and Betula maximowicziana. Compare transverse, radial and tangential sections and check each specimen record.

Describe your observations

- Which plane is shown, and where is the scale?
- Are large vessels concentrated in bands or distributed more evenly?
- Where might a ring boundary lie? Which anatomical changes support your proposal?
- What cannot be decided from a single image?

Wood anatomy varies among species and specimens. Do not identify a species or a ring boundary from one colour or pore alone; compare other sections, magnifications and specimen records. Check the provider's conditions before reusing an external image.